

Forkable Sandboxes: Five Workflows, One Loop

Workflow design connects useful starting state to evaluated outcomes and retained artifacts.

Prepared state



Explore alternatives



Evaluate concrete outcome



Keep verified artifact

DOMAIN EXAMPLES

Software factory

43:44 / \$10.33

Two redirect hosts; six-file allowlist plan.

Airflow + Docker Sandboxes; model through Databricks.

Suite and review ran; the direct adoption test remained absent.

[Run record](#)

Genomics assays

6 of 8 combinations

MCF-7 assays: greater killing with SRC-3 inhibitor SI-12.

Input: >120,000 guides; 19,050 genes. Rank, then assay.

Wet-lab replication; source identity.

[Paper and methods](#)

Physics continuation

135,445 source orbits

Ori Chamo + Eliaz: 26 difficult links checked bidirectionally.

Predictor + shooting correction connected the sampled component.

Checked numerical claims; preliminary preprint.

[Preprint](#) [Atlas](#)

AI experiments

126 attempts / 23 kept

Autoresearch: sequential search, 5 min TRAIN per attempt.

Validation bits per byte: 0.997900 → 0.969686.

AlphaEvolve: 0.7% of Google compute capacity recovered through algorithm search.

[Program](#) [Run](#) [AlphaEvolve](#)

Security campaigns

AFL++ persistent: 10-20x

1,000-input worker loop with per-input reset.

CyberGym / OpenHands + GPT-5: 431 projects; 1,748 executables.

56 candidate crashes; 22 confirmed zero-days.

Persistent mode reuses process state; the campaign tests vulnerability outcomes.

[AFL++](#) [CyberGym](#)

EXECUTION DESIGN Worktrees for source; checkpoints for live state; persistent process reset for repeated inputs.

Domain-specific outcomes retain their evaluation unit. Choose execution reuse to match the state required.

Yossi Eliaz • Cambridge SRG • Source links appear inside each case.